DEVILS DEN

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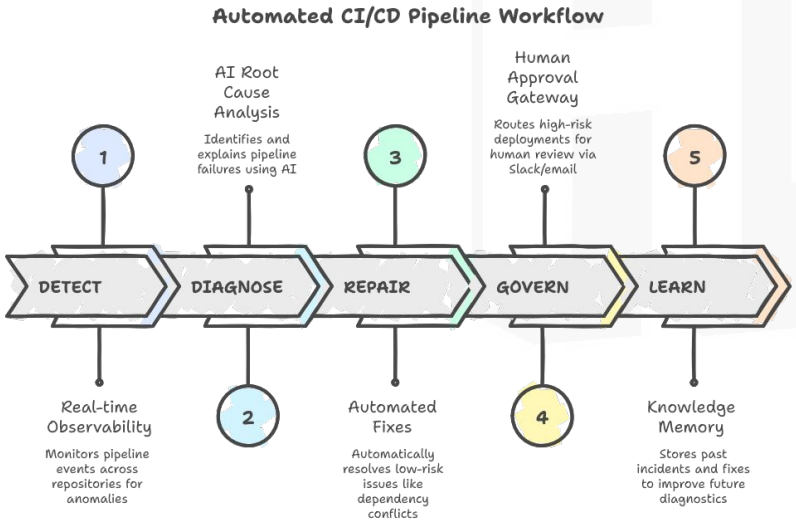
Agentic CI/CD Repair & Intelligent Release Orchestration

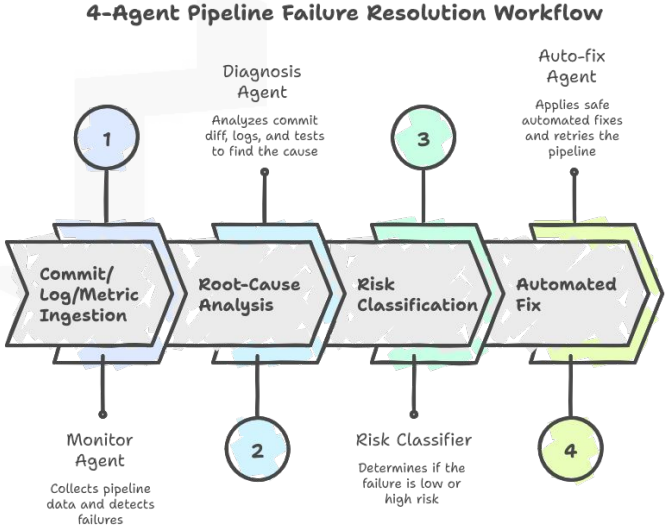
T1PS1

Modern software teams rely on complex CI/CD pipelines to deploy code frequently, but diagnosing pipeline failures often requires engineers to switch between multiple tools, such as version control, logs, test reports, and monitoring dashboards. As development velocity increases and AI-generated code becomes more common, the number of pipeline failures rises, leading to greater debugging effort and growing technical debt. At the same time, automation in DevOps pipelines can introduce security and reliability risks if autonomous systems operate with elevated permissions. The challenge is to build an Agentic CI/CD Orchestration System where autonomous agents monitor pipeline health, diagnose failures using contextual signals from commits, logs, and tests, automatically fix common issues, and enforce a human-in-the-loop governance layer for high-risk deployment decisions.

Solution Overview: **Pipeline IQ - Agentic CI/CD Manager**

A multi-agent orchestration system that monitors CI/CD pipelines in real-time, diagnoses failures using contextual signals from commits, logs, and tests then repairs common issues while enforcing human-on-the-loop governance for high-risk deployments actions. Built on event-driven architecture powered by Apache Kafka, independent AI agent process pipeline events asynchronously, enabling scalable monitoring, automated remedies and complete auditing of the fixes





Technical Stack

Frontend

- React 19
- Tailwind CSS
- WebSocket Client
- Recharts

Agentic Server

- Python 3.10-3.11
- FastAPI
- Langchain
- OpenAI GPT/Groq

Event & Messaging

- Apache Kafka
- Kafka-python
- Topic based communication agents

Pipeline Integration

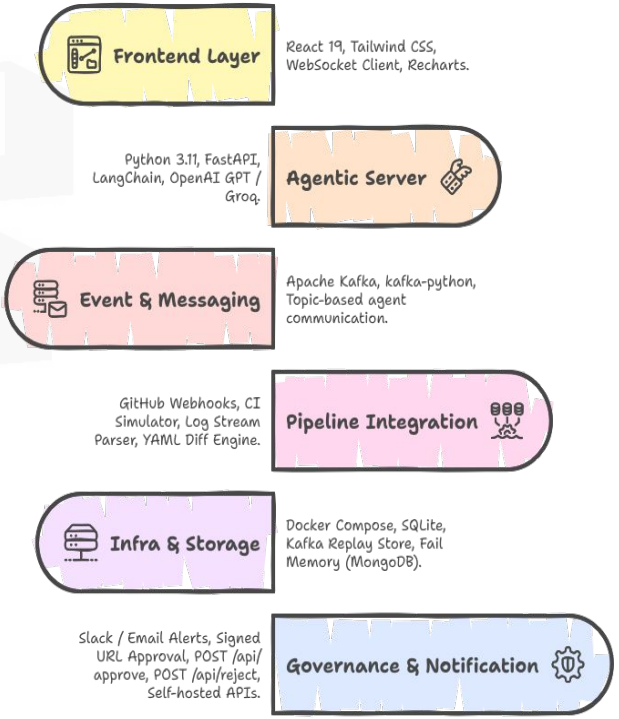
- Github webhooks
- CI Simulator
- Log Stream Parser
- YAML Engine

Infra & Storage

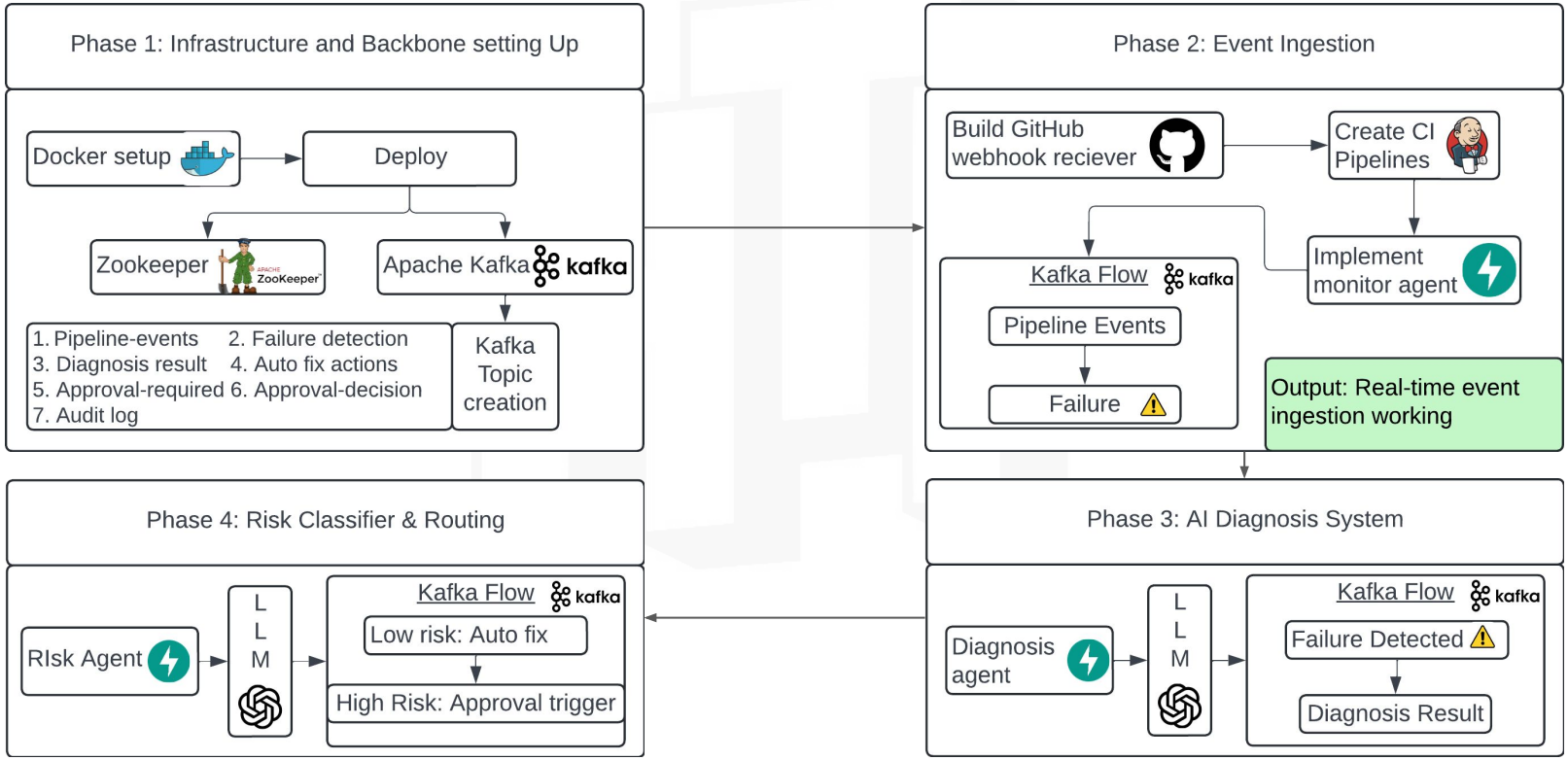
- Docker
- SQLite
- Kafka Replay Store
- Fail Memory (MongoDB)

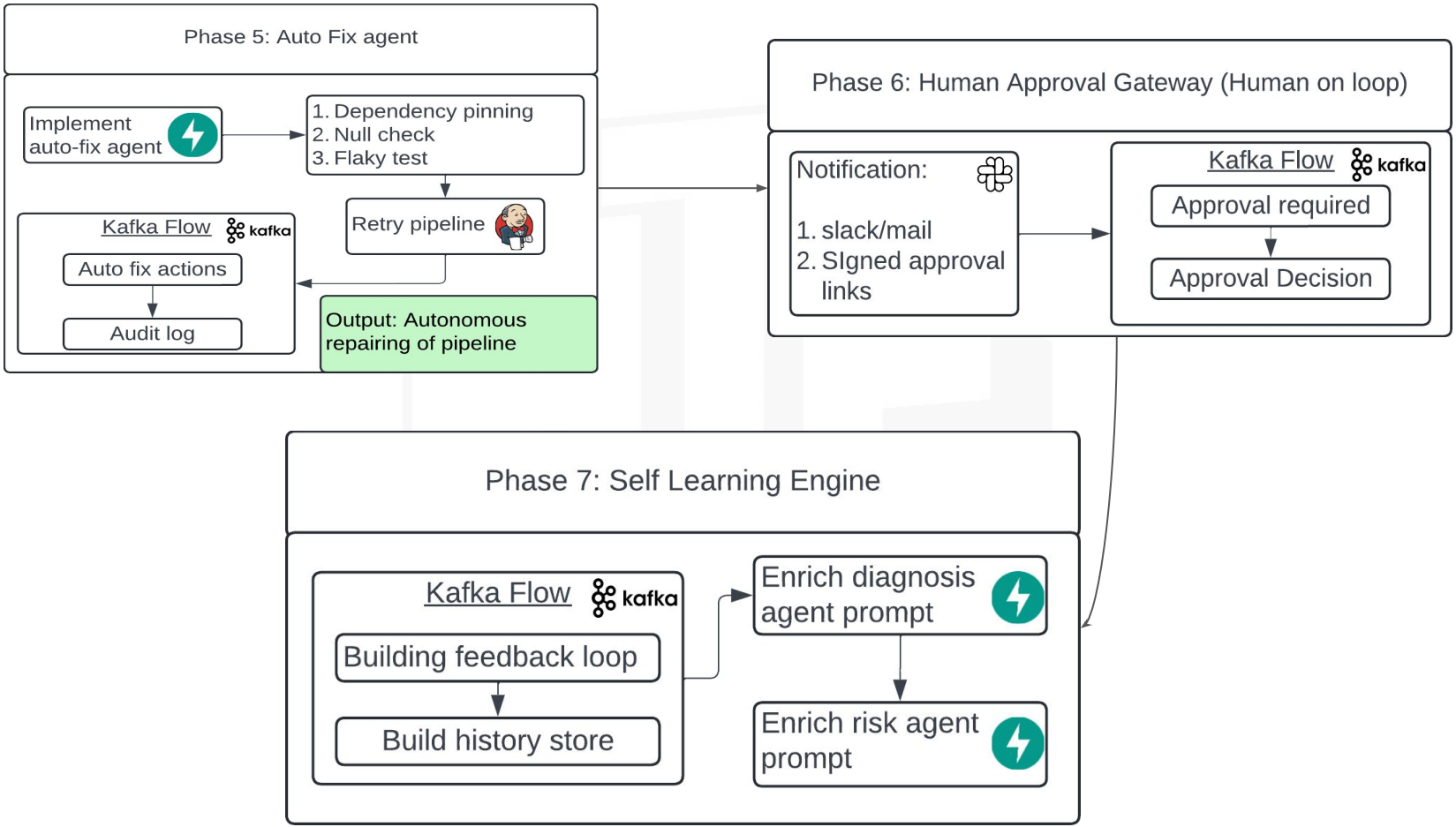
Governance & Notification

- Slack/Email
- Signed URL for approval
- Self APIs

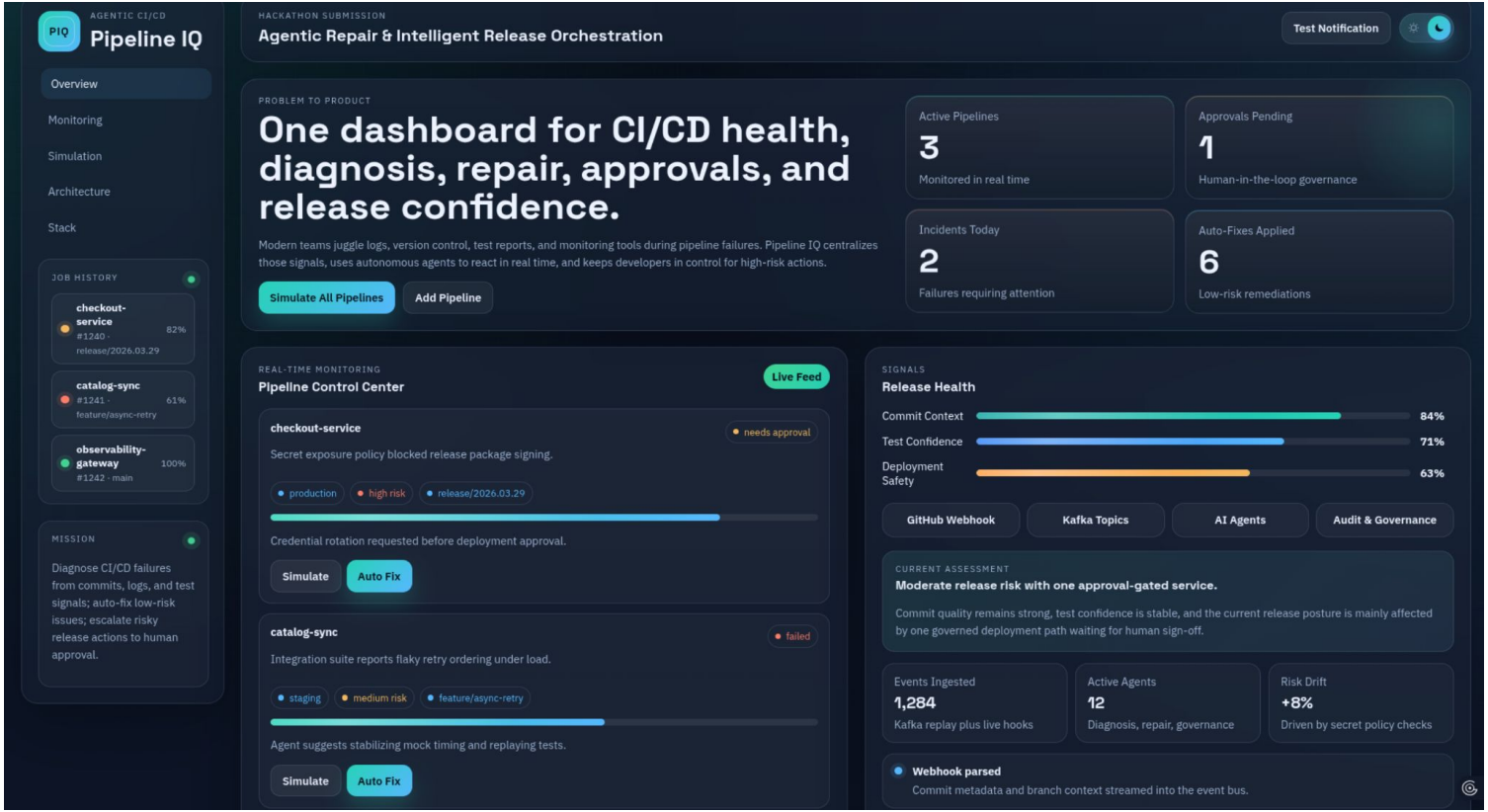


Implementation




Phase 5: Auto Fix agent
Implement auto-fix agent
1. Dependency pinning
2. Null check
3. Flaky test
Retry pipeline
Kafka Flow
kafka
Auto fix actions
Audit log
Output: Autonomous repairing of pipeline
Phase 6: Human Approval Gateway (Human on loop)
Notification:
1. slack/mail
2. SIgned approval links
Kafka Flow
kafka
Approval required
Approval Decision
Phase 7: Self Learning Engine
Kafka Flow
kafka
Building feedback loop
Build history store
Enrich diagnosis agent prompt
Enrich risk agent prompt

User Interface and Simulation Link



Deployed simulation link: https://pipeline1q-agent.vercel.app